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Research on risk analysis and precautionary measures of coal-and-gas outbursts in hydraulic tunnels

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Abstract

In order to explore the risk of coal-and-gas outbursts in hydraulic tunnels and their prevention and control measures, the No. 6 Middle Coal Seam, at the inlet section heading face of Shuidaqiao Tunnel in Guizhou Province, China, was taken as the research background. Using the testing data, the No. 6 Middle Coal Seam was determined to have a high risk of coal-and-gas outburst by both the single-index analysis method and the comprehensive *D* and *K* index analysis method. An integrated outburst precautionary measure of “advanced geological prediction + grouting cut-off wall + gas pre-drainage + grouting sealing + effectiveness verification” in a segmented and multi-cycle manner was designed and implemented. The on-site gas inspection results show that no further occurrences of gas over-limit or outburst phenomena have been observed, indicating that the above coal-and-gas outburst precautionary measure is scientific and effective. The research results can provide references for risk analysis of coal-and-gas outburst and its prevention and control in similar hydraulic tunnels passing through coal seams.

Keywords Hydraulic tunnel, Index analysis method, Coal-and-gas outburst, Risk analysis, Precautionary measure

1 Introduction

The incidence of traversing coal seams during the construction of underground engineering tunnels is not uncommon. One of the most severe dynamic disasters in such environments is the coal-and-gas outburst, which often results in the ejection of large quantities of gas and coal dust, leading to significant casualties and economic losses. It is of considerable theoretical and practical importance to systematically assess and effectively control coal-and-gas outburst risks in these tunnels, thereby minimizing construction hazards.

Coal-and-gas outburst is a dynamic disaster with extremely complex origins and numerous influencing factors. The first recorded coal-and-gas outburst occurred at Isaac Colliery in the Rouaire Coalfield, France, in 1834 [1]. Since then, many scholars have conducted extensive

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and in-depth research on its occurrence mechanisms, prediction technologies, and preventive measures, achieving numerous research results [1–23].

(1) Research on the occurrence mechanism of coal-and-gas outbursts. The coal-and-gas outburst is complex and irreproducible, making its mechanism highly complicated. The mechanisms underlying outbursts under different conditions are not yet fully understood. Most current research is based on theoretical analysis, laboratory tests, or statistical data, which only partially explain the phenomena observed during outbursts. The mechanism of coal-and-gas outburst remains hypothetical and has not yet evolved into a comprehensive theoretical system widely accepted by the academic community [2, 3]. Early scholars categorized the recognized mechanisms into four primary hypotheses according to the dominant influencing factor: the gas-dominant hypothesis [4, 5], the geostress-dominant hypothesis [6, 7], the chemical essence hypothesis [8, 9], and the comprehensive function hypothesis [10, 11].

(2) Research on the prediction technology (or risk analysis) of coal-and-gas outbursts. Major coal-producing countries around the world have developed a variety of prediction methods focused on the main factors that cause coal-and-gas outbursts, achieving effective results in engineering practice. Internationally, prediction is categorized into regional, local, and daily predictions; in China, it is divided into regional and working face predictions [12]. Various methods exist for predicting coal-and-gas outbursts, now summarized into the following categories: the index method of outburst-influencing parameters [13, 14], the geological statistical method [15, 16], the geophysical method [17, 18], and the mathematical prediction method [19, 20]. Despite numerous advancements in the field of coal-and-gas outburst prediction technology, outbursts have not been completely eradicated. Single-index predictions often fail to fully reflect the outburst risk, occasionally leading to inaccurate forecasts. A comprehensive judgment based on multiple indicators can address the shortcomings of single-index predictions and is considered likely to enhance prediction accuracy [21, 22].

(3) Research on the precautionary measures for coal-and-gas outbursts. Various scholars have proposed their views on precautionary measures for coal-and-gas outbursts, which have now essentially formed measures applicable to the safety of coal mining. These measures are categorized into regional anti-outburst measures and local anti-outburst measures, creating a double four-in-one (here, “double” refers to regional and local comprehensive outburst prevention measures. Among them, regional comprehensive outburst prevention measures include regional outburst danger prediction, regional outburst prevention measures, regional measure effect verification,

and regional verification. Local comprehensive outburst prevention measures include working face outburst danger prediction, working face outburst prevention measures, working face measure effect verification, and safety protection measures. The anti-outburst technical system is based on the principle that regional comprehensive anti-outburst measures should be implemented as the priority, with local anti-outburst measures serving as a necessary supplement [23, 24]. At a joint academic exchange meeting of the Coal Society in 2018, Huang [25] provided a relatively objective assessment of the development trends in coal-and-gas outburst research in China. The author suggested that future developments should focus on enhancing the understanding of the outburst mechanism, improving the simplicity and accuracy of outburst risk prediction, and diligently implementing the double four-in-one anti-outburst measure.

Despite extensive research on the mechanisms and preventive measures for coal-and-gas outbursts, the focus has been primarily on coal mines. Breakthroughs in gas disaster prevention in the field of hydropower tunnels have been limited. It is important to note that gas accidents in hydropower tunnel excavation projects differ from mine gas disasters in terms of their mechanisms, development processes, and characteristics. However, due to the extremely limited research on related mechanisms, control technologies, and applicable standards, current standards and specifications largely adhere to those of the coal industry [26, 27]. Therefore, it is urgent to conduct research on gas disaster prevention and control technologies tailored to the specific needs of hydropower tunnel projects.

Based on the background provided, an issue of coal-and-gas outburst was encountered in the inlet section of the Shuidaqiao Tunnel in Guizhou Province while passing through the No. 6 Middle Coal Seam. For this project, tests on the main index parameters of coal and gas were conducted. These tests facilitated a systematic analysis of the outburst risk associated with coal and gas in the No. 6 Middle Coal Seam using both single-index and comprehensive-index prediction methods. Subsequently, targeted precautionary measures for managing coal-and-gas outburst in the coal seam were developed and implemented to ensure safe and normal construction of the Shuidaqiao Tunnel through this coal seam. The research results are expected to enrich the theoretical foundation and practical experience in the field of risk analysis and precautionary measure development for coal-and-gas outburst in hydraulic coal-crossing tunnels.

2 Project overview

The Shuidaqiao Tunnel is part of the construction sections of the North Canal forepart in the Jiayan Water Conservancy Hub Project located in Guizhou Province.

The tunnel spans approximately 20 km and is divided into two sections: Section 1 and Section 2 of the North Canal. The high-gas section of the Shuidaqiao Tunnel in Section 1 extends from the tunnel inlet to the section marked K1+410, and the coal-and-gas-outburst prone section lies between the sections marked K1+410 and K1+600. Strata revealed at the heading face belong to the Upper Permian Longtan Formation (P_{31}) of the Carboniferous System, consisting of interbeds of mudstone, sandstone, and coal seams. The surrounding rock is classified as Class V. Before reaching the No. 6 Middle Coal Seam, multiple coal seams had been encountered in Section 1 of the North Canal; their dip directions are nearly parallel to the tunnel direction. The thickness of these coal seams varies, and they comprise high-quality anthracite.

At 9 a.m. on July 3, 2017, a gas inspector in Section 1 was conducting routine manual gas detection at the heading face marked K1+410 when a sudden outburst of coal dust occurred, as shown in Fig. 1, resulting in abrasions to the inspector's legs. In September 2017, a professional organization evaluated the outburst risk of the No. 6 Middle Coal Seam and concluded the following: (1) The gas dynamic phenomenon observed qualifies as a coal-and-gas outburst. (2) The No. 6 Middle Coal Seam is prone to coal-and-gas outbursts, thereby classifying the Shuidaqiao Tunnel as a coal-and-gas-outburst tunnel.

3 Test and risk analysis of outburst

Despite the classification of the No. 6 Middle Coal Seam as outburst-prone, it remains highly necessary to conduct more professional and multifaceted quantitative experimental analyses to determine the degree of outburst risk. This will ensure the relevance and accuracy of the outburst precaution plan.

3.1 Basic situation analysis

According to site investigation data, the coal-bearing strata in the Shuidaqiao Tunnel area range from approximately 201.02 m to 233.17 m in thickness, with an average thickness of 212.13 m, containing 12 coal seams. These seams are numbered from top to bottom as 6-upper, 6-middle, 6-lower, 7, 10, 16, 23, 26, 31, 33, 34, and 35. Of these, four are mineable coal seams: 6-middle, 16, 23, and 33, while the others are sporadically mineable coal seams. The minimum thickness of the No. 6 Middle Coal Seam is 0.69 m, and the maximum thickness is 6.13 m. According to previous field investigations, the No. 6 Middle Coal Seam ranges from being easily self-igniting to not being easily self-igniting, and it does not present a coal dust explosion hazard. The normal gas composition includes a CH_4 content of more than 80%, and a CO_2 content of less than 20%. This classifies it within the methane zone. The normal gas concentration is 0.2%, and the absolute gas volume is $5 \text{ m}^3/\text{min}$.



Fig. 1 The coal seam revealed at the heading face of Shuidaqiao Tunnel inlet section

Parameters of the No. 6 Middle Coal Seam and its gases will be measured through in-situ and laboratory experiments. These will be compared with the critical values specified in the Chinese Identification Regulation for Coal and Gas Outburst Mines (AQ1024-2006) [28] (this Regulation is tailored to regional climate, geological, and construction conditions in China, supplementing international codes with practical design and construction guidelines for the East Asian region) to determine whether the coal seam poses a potential outburst risk.

3.2 Gas pressure and content

(1) Gas pressure

The M-II type gas pressure measuring instrument was used, and three test holes were arranged to directly measure the gas pressure of the No. 6 Middle Coal Seam. The results are shown in Table 1.

According to the Regulation, the maximum gas pressure measured at each point was taken as the representative pressure for that coal seam. Therefore, the gas pressure of the No. 6 Middle Coal Seam was $P = 0.84$ MPa.

(2) Gas content

(a) Site desorption

In this study, a 2-h desorption test was conducted on site in accordance with the procedures specified in the MT/T77-94 standard for coal seam gas measurement. The gas samples were then sent to the laboratory for residual gas content analysis. The results are presented in Table 2.

Based on the measurements, the gas content of the No. 6 Middle Coal Seam ranged from 15.50 to 25.83 mL/g, with an average content of 21.45 mL/g across the entire area. These values indicate that the seam is a high-gas coal seam.

The gas gradient indicated that for every 9.96 m of burial depth increase, the gas content increased by 1 mL/g on a dry ash-free (daf) basis.

The gas content gradient indicated that for every 100-m increase in burial depth, the gas content increased by 10.04 mL/g (daf).

(b) Sampling test

Coal samples were taken from the No. 1 and No. 2 drill holes, and the gas contents were measured directly on site with the DGC gas content direct measuring device. The original gas content of the No. 6 Middle Coal Seam was measured at 14.778 m³/t.

3.3 Gas outburst parameters

(1) Coal type of damage

A representative coal sample was taken from each of the holes, as indicated in Fig. 2.

Upon examination, the coal sample displayed a primarily semi-bright and semi-dark luster, retaining its layered



Fig. 2 Coal sample of the No. 6 Middle Coal Seam

Table 1 Parameters of pressure-measuring drill holes and test results

Coal seam	Hole number	Angle with tunnel	Dip angle	Hole depth	Hole diameter	Sealing length	Gas pressure	Notes
No. 6 Middle	1#	0°	41°	12.8 m	75 mm	8.5 m	0.84 MPa	Hissing sounds and jetting
	2#	25°	2°	37.3 m	75 mm	9.1 m	0.80 MPa	
	3#	−25°	2°	37.3 m	75 mm	9.5 m	0.82 MPa	

Table 2 Gas composition and content analysis of the coal seam

Coal seam	Hole number	Floor elevation (m)	Natural gas composition (%)				Gas content (mL/g)				
			Air-free basis				CO ₂	CH ₄	C ₂ H ₆	CO	Total
			N ₂	CO ₂	CH ₄	C ₂ H ₆					
No. 6 Middle	202	1311.05	4.69	0.37	94.72	0.23	0.07	16.39	0.06	0	16.52
	203	1198.43	10.68	0.07	89.19	0.07	0.07	24.69	0.07	0	24.83
	301	1345.58	5.15	0.45	94.31	0.09	0.04	15.42	0.04	0	15.50
	303	1122.68	7.97	0.18	91.78	0.06	0.09	22.36	0.05	0	22.49

structure. It appeared relatively orderly, with distinct bands, irregular block shapes, and numerous corners and secondary joints. The joints were irregular and discontinuous. The coal sample was friable, breaking easily into small fragments that can be manually ground into powder. According to the regulation on coal damage classification, the No. 6 Middle Coal Seam was classified as Type IV, i.e., crushed coal.

(2) Consistency coefficient

The coal strength was characterized by the consistency coefficient f , determined using the standard drop-hammer crushing method:

- Small coal lumps (20–30 mm) were selected from the samples and divided into five 50-g portions, each of which was placed in a measuring cylinder for the crushing test.
- The drop hammer fell freely from a height of 600 mm and stroke each portion three times.
- After all five portions were crushed, they were poured onto a 0.5-mm mesh screen. The undersize material was collected in a measuring cylinder to determine the height h of the coal powder.

The value of f can be calculated using the following formula [29].

$$f_{20-30} = 20n/h \quad (1)$$

where f_{20-30} is the measured consistency coefficient of the coal sample with a particle size of 20–30 mm; n is the number of drop-hammer impacts; h is the height of the coal powder measured in the measuring cylinder (mm).

The coal sample of the coal seam is soft coal and its particle size cannot reach 20–30 mm. So, coal sample with a particle size of 1–3 mm can be taken for measurement, and f can be converted as follows:

$$\begin{cases} f_{20-30} = 1.57f_{1-3} - 0.14 & \text{When } f_{1-3} > 0.25; \\ f_{20-30} = f_{1-3} & \text{When } f_{1-3} \leq 0.25 \end{cases} \quad (2)$$

where f_{1-3} is the consistency coefficient of coal sample with a particle size of 1–3 mm.

The test results are listed in Table 3.

According to the Regulation, the consistency coefficient of coal in the same seam should be taken as the minimum value among all measurement points. Therefore, the consistency coefficient of the No. 6 Middle Coal Seam is 0.278.

(3) Initial speed of methane emission

The initial speed of methane emission (ΔP) in the coal sample from the No. 6 Middle Coal Seam was measured using the WT-1 gas emission speed measuring instrument. The test results are summarized in Table 4.

According to the Regulation, the initial speed of methane emission in the same coal seam should be taken as the maximum value among all measurement points. Therefore, the initial speed of methane emission in the No.6 Middle Coal Seam is 41.0 mmHg.

3.4 Outburst risk analysis

Currently, research on risk analysis methods for coal-and-gas outbursts in hydropower tunnels is extremely scarce, and the indicator-based method remains one of the most common approaches. In the following analysis, the risk of coal-and-gas outburst in the No. 6 Middle Coal Seam of the Shuidaiao Tunnel was evaluated using both the single-indicator and comprehensive-indicator method.

(1) Single-indicator analysis of outburst risk

(a) During probe drilling prior to construction in the Shuidaiao Tunnel, premonitory signs of outburst, including drill sticking, rod jamming, and spouting from boreholes, were observed. On July 3, 2017, a gas dynamic phenomenon consistent with the characteristics specified in the Regulation occurred in the No. 6 Middle Coal Seam. Therefore, this coal seam in the Shuidaiao Tunnel posed a risk of outburst.

(b) The damage type, gas pressure, consistency coefficient, and initial gas emission velocity of the No. 6 Middle Coal Seam of the Shuidaiao Tunnel all reached or exceeded the critical values specified for each single indicator in the Regulation (as shown in Table 5). Therefore, the No. 6 Middle Coal Seam of Shuidaiao Tunnel was considered at risk of outburst.

The measured prediction indicators for outburst risk in the No. 6 Middle Coal Seam all exceeded the critical

Table 4 Determination of initial gas emission speed in the No. 6 Middle Coal Seam

Sampling location	Number of test	Initial speed of methane emission ΔP (mmHg)	Maximum value of ΔP (mmHg)
Hole 1#	1	33.0	41.0
	2	32.8	
	3	41.0	

Table 3 Consistency coefficient test of the coal seam

Sampling location	Number of test	Sampling particle size	Number of impacts	Height of coal powder	f_{20-30}	Minimum value of f_{20-30}
Hole 1#	1	1–3 mm	3	225 mm	0.278	0.278
	2	1–3 mm	3	220 mm	0.288	

Table 5 Comparison of single indicators for outburst identification in the No. 6 Middle Coal Seam

Comparison value	Destruction type	Initial speed of methane emission ΔP	Gas content	Consistency coefficient f	Gas pressure P
Critical value	III, IV, V	≥ 10 mmHg	≥ 8 m ³ /t	≤ 0.5	≥ 0.74 MPa
Measured value	IV	41.0 mmHg	14.778 m ³ /t	0.278	0.84
Whether it exceeds the critical value	Yes	Yes	Yes	Yes	Yes

values and fell within the dangerous ranges specified in the China's Regional Prediction Methods for Coal and Gas Outburst Danger (GB/T 25216—2024) [30] and the Regulations on Prevention and Control of Coal and Gas Outburst (2019) [31]. As indicated by the single-indicator analysis, the No. 6 Middle Coal Seam of the Shuidaiao Tunnel was predicted to be at outburst risk.

(2) Comprehensive-indicator analysis of outburst risk

Only one aspect of outburst behavior can be reflected by single-indicator analysis, and it is rarely used alone. However, the roles of various individual factors in the outburst process can be comprehensively reflected by comprehensive-indicator analysis; thus it is a commonly used prediction method for coal-and-gas outburst hazard. The D and K combined-index method is one such important method. According to the Regulations on the Prevention and Control of Coal and Gas Outbursts (2019), when using D and K indices for outburst-hazard prediction, if either of the two indices exceeds the standard, the area is considered prone to outburst. Only when neither of these two indices exceeds the standard is the working face considered safe.

According to relevant specifications and previous studies [31, 32], the D and K indexes are composite indexes, each consisting of single indicators. Their values are calculated using the following formulas.

$$\begin{cases} D = (0.0075H/f - 3) \times (P - 0.74) \\ K = \Delta P/f \end{cases} \quad (3)$$

where H denotes the depth of the coal seam (m).

For the No. 6 Middle Coal Seam of the Shuidaiao Tunnel, the average depth from the surface was $H = 212.13$ m. Substituting the relevant values into Equation (3), we obtained the comprehensive indexes D and K as 0.272 and 147.48, respectively. Both values exceeded the critical thresholds specified in the Regulation, as listed in Table 6 [33].

The comprehensive indexes D and K for outburst risk in the No. 6 Middle Coal Seam exceeded the critical

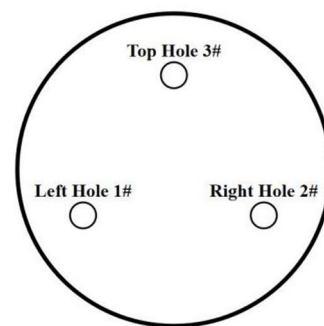
values specified in China's Regulations on Prevention and Control of Coal and Gas Outburst (2019). In conclusion, the comprehensive-indicator analysis indicates that the No. 6 Middle Coal Seam of the Shuidaiao Tunnel is at outburst risk.

4 Outburst precaution plan

Based on the gas parameter measurements and outburst risk analysis presented in Section 3, the No. 6 Middle Coal Seam was classified as outburst-prone. To prevent gas accidents and ensure construction safety and structural stability in the inlet section of the Shuidaiao Tunnel, comprehensive gas control measures must be implemented in the outburst-prone coal seam sections. Following treatment, an effectiveness inspection must be performed to confirm that the working face is free of outburst risk before resuming normal excavation. The detailed plan is outlined as follows.

4.1 Advanced geological forecasting

Advanced horizontal geological drilling holes were installed at the working face (Fig. 3). These drilling holes had a diameter of 94 mm, an inclination of 0°, and a depth of 93.7 m. The type of rock mass is roughly determined based on the drilling fluid returns and drill cuttings from these holes. The stratigraphic columns for the three drilling holes are presented in Fig. 4.

**Fig. 3** Layout of advanced drilling**Table 6** Reference critical values of D and K

Critical value in the Regulation	Comprehensive indicator D	Comprehensive indicator K	
		Anthracite	Other coal types
	≥ 0.25	≥ 20	≥ 15
Values of No. 6 Middle Coal Seam	0.272	147.48	/

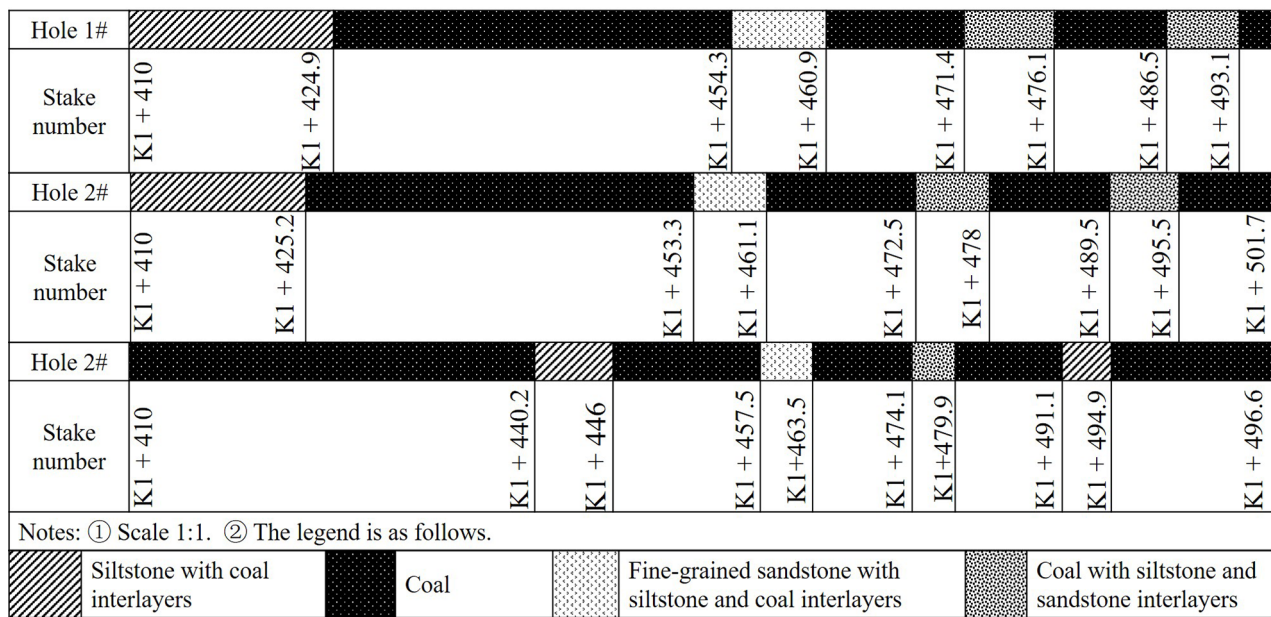


Fig. 4 Columnar diagram of advanced horizontal drilling hole at 1:1 scale

According to advanced exploration data, the coal seam in the 100-m section ahead of the tunnel occurred frequently, interbedded with thin siltstone and fine-grained sandstone layers. Due to the gently inclined strata, the coal seam persisted over a greater distance. The coal-bearing strata were expected to be encountered at chainage K1 + 600. During the drilling of the three advanced geological holes, rod bouncing occurred at various depths. Upon withdrawing the drill rods, gas emanated from the rods. The gas concentration in the holes exceeded 40%, with a maximum of 56%. Additionally, during the drilling process, continuous occurrences of hole spouting and frequent sounds of small explosions (coal bursts) were observed within the holes.

Based on the analysis of advanced exploration drilling and compiled data, the coal seam exposed at the tunnel inlet (chainage K1 + 405) of the Shuidaqiao Tunnel was identified as the No. 6 Middle Coal Seam. The tunnel is projected to exit the coal-bearing strata near chainage K1 + 600 and enter the P_{3c} limestone strata. In the absence of anomalous structural conditions, the remaining length of the tunnel to be excavated through the coal-bearing strata was estimated at approximately 200 m.

4.2 Establishment of full-section grouting cut-off wall

The excavation at the inlet section of the Shuidaqiao Tunnel consisted solely of tunnel excavation and did not involve seam recovery or mining operations. The main focus was outburst prevention at the working face. Currently, there are no design or construction specifications for tunnels passing through coal seams in the water conservancy engineering industry. Drawing on relevant

design standards from the coal mining, highway, and railway sectors, and considering the characteristics of water conservancy projects, the design incorporated factors such as construction scheduling, cost control, and construction safety. The design was based on local outburst prevention measures, combining gas drainage and sealing. It adopted a gas control strategy using a segmented multi-cycle approach, namely, an integrated outburst prevention system of “advanced geological prediction + grouting cut-off wall + gas pre-drainage + grouting sealing + effectiveness verification,” implemented in segments of 25 m with a 5 m overlap between adjacent segments.

Simultaneously, to ensure construction safety and grouting quality, a full-section grouting cut-off wall was installed at the tunnel face. The wall was constructed from C20 concrete with a thickness of 1.5 m. For anchoring, M20 mortar and C25 bolts were used, connected to and integrated with the initial support. Each bolt was 3 m long and installed at 0.5 m spacing.

4.3 Design of outburst prevention measure

From a technical perspective, the outburst-prevention measures at the heading face address three core aspects: outburst risk prediction, preventive measure design, and verification of their effectiveness.

(1) Outburst risk prediction

In each excavation cycle of the outburst-prone working face, the outburst risk was predicted. The outburst risk prediction was performed using the drill cuttings indices method, specifically the gas desorption index of drill cuttings (K_1) and the drill cuttings quantity (S). For

the exposed No. 6 Middle Coal Seam, three prediction holes were arranged along the soft layer of the coal seam, with a hole diameter of 42 mm and a depth of 30 m. Only when the gas desorption index K_1 of drill cuttings was less than the critical value [$0.5 \text{ mL}/(\text{g} \times \text{min}^{1/2})$ for dry coal sample and $0.4 \text{ mL}/(\text{g} \times \text{min}^{1/2})$ for wet coal sample], the quantity of drill cuttings S was less than 6 kg/m , and no other abnormal conditions were found, was the prediction result considered free of outburst risk. In this case, a 5-m safety section was left to allow the excavation of 25 m before the next round of prediction. Otherwise, the working face was considered to be at risk of outburst, and local outburst prevention measures must be implemented. Excavation work can only proceed after the prediction showed no outburst risk.

(2) Prevention measure design

(a) Pre-drainage of gas

The tunnel cross-section was circular with a radius of $r=2.7 \text{ m}$, and the excavation cross-section had a radius $a=3.50 \text{ m}$. In the absence of measured drilling drainage radius data, the bottom spacing of gas drainage drilling holes was designed to be 5 m. There were five gas drainage holes per round, including the central grouting hole ④ and four grouting holes ②. Grouting hole ② were alternated in use, and their drainage depth matched the grouting depth. The layout is illustrated in Fig. 5.

Gas drainage termination was determined by two criteria: gas concentration less than 10% and gas pressure less than 0.2 MPa. If these criteria are not met, the requirement was to continue gas drainage for an additional 3 days post-grouting.

(b) Grouting to solidify the coal seam

After the completion of gas drainage, the coal seam ahead of the tunnel was consolidated via grouting to enhance its cohesion coefficient and seal gas around the tunnel excavation working face. Grouting was carried out according to the principle of working from the inside out,

in the sequence ④-①-②-③ as shown in Fig. 6. The details of the grouting holes are as follows:

Grouting hole ①: Located 1.85 m from the edge of the heading face, these were evenly arranged in the circumferential direction, with 8 holes per round, a hole diameter of 94 mm, a hole depth of 37.7 m, and inclined outward at 9° .

Grouting hole (gas drainage and grouting hole) ②: Positioned 0.6 m from the edge of the heading face, evenly arranged in the circumferential direction, with 8 holes per round, a hole diameter of 94 mm, a hole depth of 30.0 m, and inclined outward at 12° .

Grouting hole ③: Located 0.6 m from the edge of the heading face, evenly arranged alternately with grouting hole ②, with 8 holes per round, discontinued in the second round, a hole diameter of 94 mm, a hole depth of 12.5 m, and inclined outward at 24° .

Grouting hole ④: Constituted by 1 hole per round, with a hole diameter of 94 mm, a hole depth of 35.0 m, and arranged perpendicular to the heading face.

Inspection hole ⑤: Positioned 1.2 m from the edge of the heading face, evenly arranged in the circumferential direction, with 4 holes per round, a hole diameter of 42 mm, a hole depth of 25.0 m, and arranged perpendicular to the heading face.

Grouting hole ④ and selected holes ② also served as gas drainage holes in the early stage. The holes were drilled first and then grouted. After drilling was completed, the interval before grouting must be at least 3 days.

The grouting material was ordinary Portland cement slurry (grade ≥ 42.5) with a water-cement ratio of 0.6. If the slurry consumption exceeded 1000 kg/m^3 , a sodium silicate solution can be added, with a volume ratio of cement slurry to sodium silicate of 1:0.7. Compaction grouting was adopted to reinforce the surrounding rock mass and seal gas leakage.

Grouting was conducted following the principle of segmental grouting from the inside out. Each 5 m section

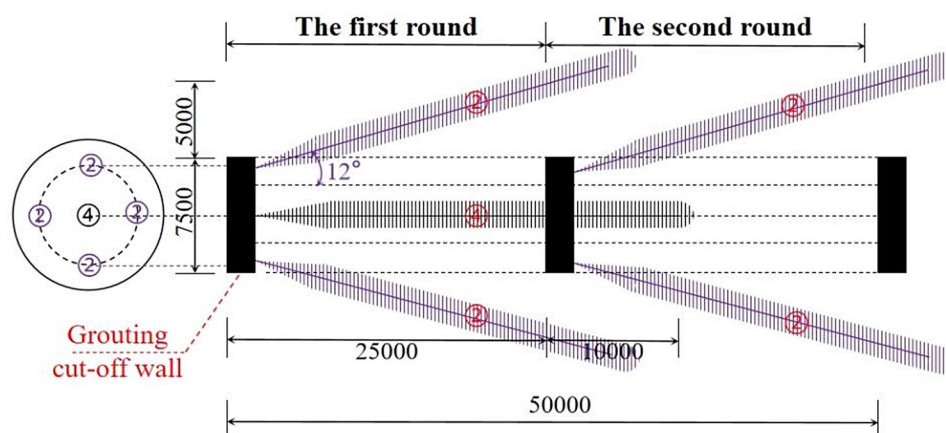


Fig. 5 Plan view of gas pre-drainage drilling holes

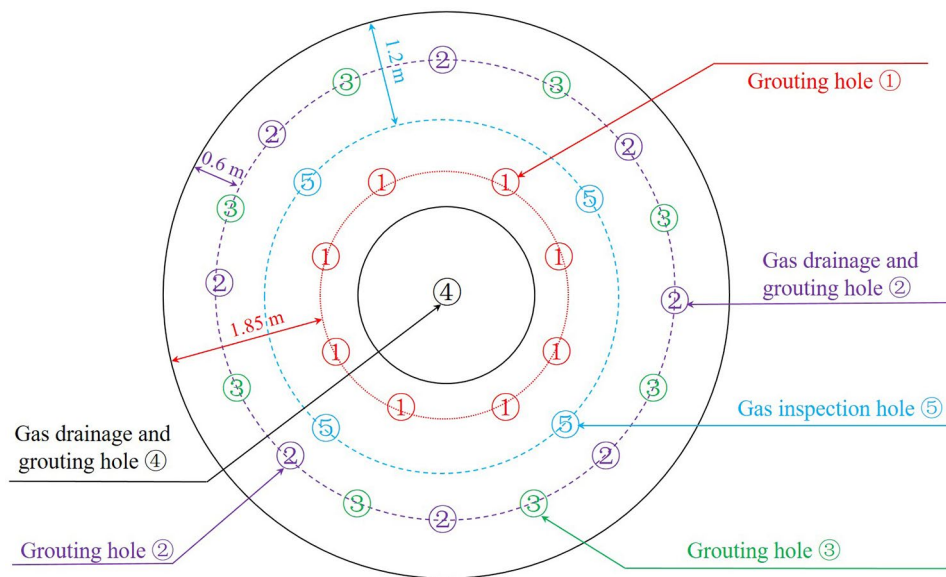


Fig. 6 Plan view of gas pre-drainage drilling, grouting, and inspection holes

was defined as one segment. The grouting pressure for the first segment was 0.3 MPa, for the second segment 0.8 MPa, and for the third segment and beyond 1.5 MPa. The grouting pressure can be appropriately adjusted based on the conditions observed in the grouting cut-off wall throughout the process, provided that the adjustments avoid inducing cracks in the initial support or the grout curtain.

The completion of grouting was determined when the injection rate fell to no more than 1 L/min, and grouting continued for an additional 30 min to ensure full consolidation.

(3) Effect verification of the measure

After completing gas drainage and consolidation grouting, the effectiveness of the measures was inspected along

the grouting cut-off wall. The drill cuttings gas desorption index was measured through inspection hole ⑤. The inspection results were considered satisfactory and free of outburst risk only if K_1 was less than the previously mentioned critical value, S was less than 6 kg/m, and no other abnormal conditions were detected. If these conditions were not met, the working face was deemed at risk of outburst, and local outburst prevention measures must be reinstated. This process was repeated until the inspection confirmed that there was no outburst risk.

4.4 Construction technology of prevention measure

The construction technology flow of the one-stop outburst prevention measure is presented in Fig. 7.

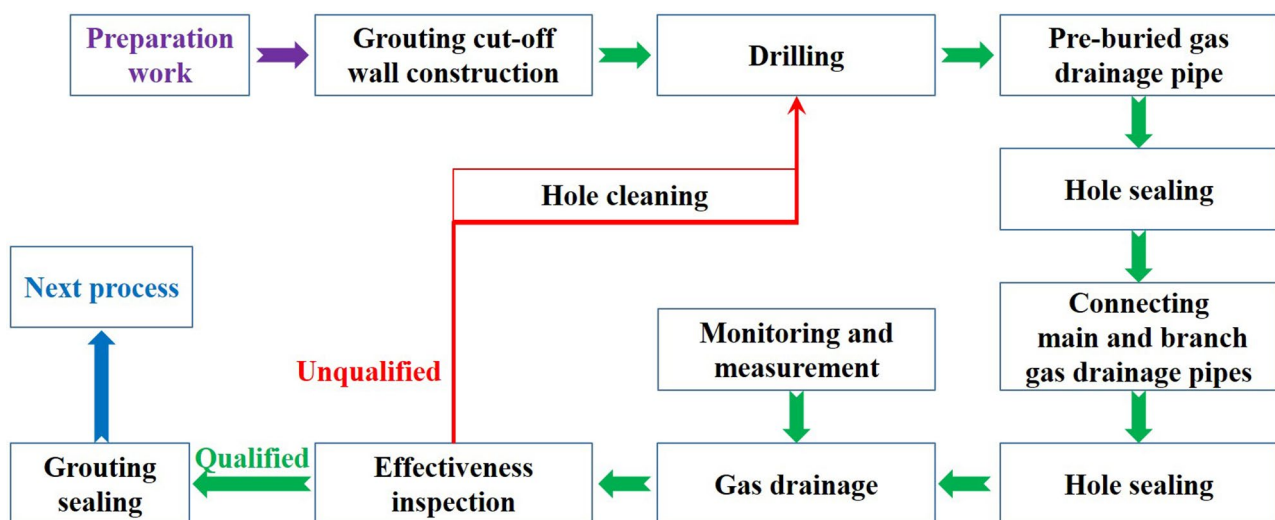


Fig. 7 Construction technology of the one-stop outburst prevention measure

4.5 Field application effect

After resuming construction at the inlet section of the Shuidaqiao Tunnel, various measures were implemented, including drilling, gas drainage, grouting and sealing, and the excavation and support of upper and lower benches (as shown in Fig. 8). Concurrently, gas detection and enhanced ventilation were carried out. The specific actions (as shown in Fig. 9) included the installation of 15 gas probes at the tunnel entrance, the heading face, and the return airway. Three optical gas detectors were purchased (one as a backup) and connected in series to the host in the tunnel entrance duty room via cables, linked to a display screen. An over-limit gas concentration alarm system was established (with automatic alarms triggered at concentrations exceeding 0.5% and power cut-off at concentrations exceeding 0.75%). Manual detection and the intelligent gas detection system were used concurrently for dual control, ensuring that on-site construction safety remained within acceptable limits. Additionally, two ventilators with a total power of 370 kW were set up for 24-h continuous ventilation, and three explosion-proof local ventilators for mines were acquired to enhance ventilation in the tunnel.

The detection results indicated that the highest gas concentration at the heading face was 0.46%, which was below the 0.5% threshold (according to the Coal-and-Gas Over-Limit Management System of Guizhou Province, a gas concentration of 0.5% or greater is classified as a

general gas over-limit). Thus, no further instances of gas over-limit or outburst phenomena were observed until the tunnel breakthrough. This confirms that the previously designed one-stop gas prevention measure is scientific and feasible, capable of ensuring normal and safe construction of the project, and can serve as an exemplary case in the field of coal and gas outburst control during the construction of water conservancy tunnels through coal seams. Furthermore, the integrated outburst prevention measure—"advanced geological prediction, grouting cut-off wall, gas pre-drainage, grouting sealing, and effect verification"—implemented in a segmented multi-cycle manner and tailored to this project, has been successfully recognized as a provincial-level construction method in Guizhou for the years 2020–2021. This method is suitable for high gas control in hydraulic tunnels and has significant potential for widespread adoption and practical application value.

5 Discussion

In this study, we present an integrated outburst prevention approach that combines "advanced geological prediction, grouting cut-off wall, gas pre-drainage, grouting sealing, and effectiveness verification" in a segmented multi-cycle manner, tailored to the construction environment of hydraulic tunnels. It is important to note that, compared to typical coal mine roadway outburst prevention measures, the implementation of outburst prevention in



Fig. 8 The on-site implementation of gas prevention and control measures

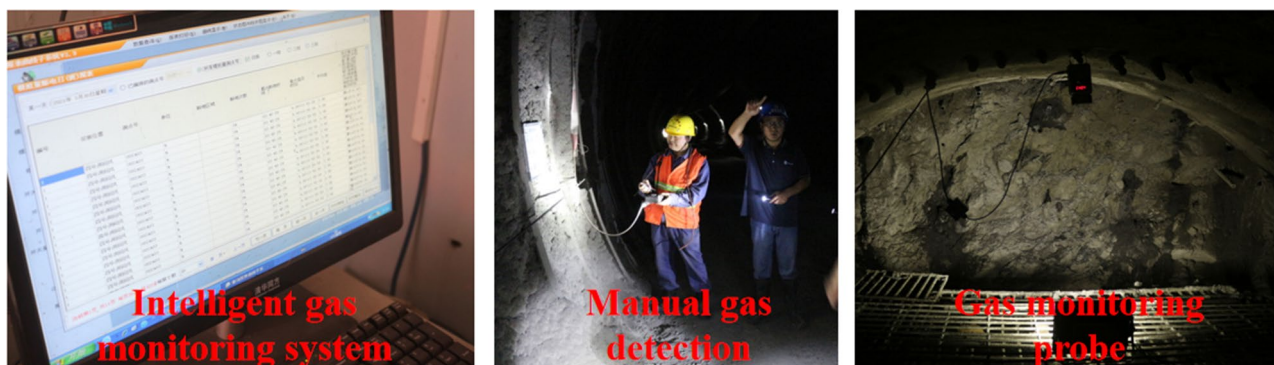


Fig. 9 The detection system and on-site implementation of gas

the Shuidaiao hydraulic tunnel offers advantages: easier transportation of equipment and materials, more efficient personnel coordination, more stable load-bearing structures, and larger operational areas. However, challenges such as limited availability of specialized outburst prevention equipment, constrained construction schedules, elevated waterproofing requirements, and insufficient technical expertise also exist. By thoroughly considering the strengths and weaknesses of coal-and-gas outburst prevention and treatment in hydraulic tunnels, the measures proposed in this study are grounded in systematic gas testing data and quantitative outburst risk analysis, and are closely aligned with the actual construction conditions of the Shuidaiao Tunnel. This alignment is precisely what enables the successful application of these measures in practice. Given the extremely limited relevant experience and standards specific to hydraulic tunnel outburst prevention, this plan still adheres to coal industry standards, while striving to shorten construction periods and reduce costs without compromising safety. However, in the practical application process, relevant parameters of the technical measures proposed in this paper should be appropriately adjusted based on thorough investigation of engineering conditions and close monitoring of gas occurrence characteristics, and must not be applied arbitrarily. The practical application results of the Shuidaiao Tunnel demonstrate that the measure was proven to be safe and reliable in this specific case, making it suitable for the construction environment of hydraulic tunnels and providing significant potential for broader application. It is worth noting that the technical measures were proposed based on the actual engineering conditions of the Shuidaiao Tunnel and might be applicable to hydraulic tunnels with similar engineering conditions, but the limitations and generalizability of these measures require further validation through additional engineering cases.

6 Conclusion

This study addresses the problem of coal-and-gas outburst risk encountered at the inlet section of the Shuidaiao Tunnel during its passage through the No. 6 Middle Coal Seam. Key technical parameters governing coal-and-gas outburst potential in the seam were systematically measured. Based on these measurements, a comprehensive outburst risk assessment of the coal seam was conducted. Integrating construction schedule, safety requirements, and cost efficiency, a one-stop gas prevention strategy—comprising “advanced geological prediction, grouting cut-off wall, gas pre-drainage, grouting sealing, and effectiveness verification”—was designed and implemented in a segmented, multi-cycle manner. Both construction practice and on-site gas monitoring confirmed that this measure effectively ensures project timeline adherence while guaranteeing safe and

uninterrupted tunnel construction. It is a scientifically sound and feasible gas control measure for hydraulic tunnels traversing coal seams, and it can be applied to the prediction and prevention of coal-and-gas outbursts in hydraulic tunnels under similar working conditions, holding significant value for promotion and practical application.

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Author contributions

Hailong Dong supervised the research and proposed the research direction. Zhengxin Zhang was responsible for the report analysis and writing of the paper. Zuotai Zhang, Hanming Yang, Wei Jing, Zheng Huang, and Zuodong Song helped write the paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

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